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# *biometryassist*: an R package to aid the design and analysis of agronomic experiments

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## Summary

We developed the *biometryassist* R package (Nielsen et al. 2026) to provide helper functions and documentation supporting the design and analysis of agronomic field experiments, and have published it on the Comprehensive R Archive Network (CRAN). Its functions cover the common steps of a field trial workflow, including generating and randomising experimental designs, fitting and checking mixed models, and presenting the results. We designed them for agricultural researchers who need robust analyses but may have limited statistical or programming experience.

The package also serves as a companion to our biometry training workshops at the Biometry Hub, Adelaide University, where we use it to reduce the cognitive load on participants so that emphasis remains on statistical concepts and interpretation rather than software mechanics. The functions and documentation deliberately align with the concepts taught, so participants can continue using the package in their own work after the workshops, alongside researchers who adopt it independently. We maintain a companion website with comprehensive documentation and usage examples at https://biometryhub.github.io/biometryassist/.

## Statement of need

Agricultural researchers are often expected to design and analyse their own experiments, yet rigorous experimental design and mixed-model analysis draw on statistical methods that fall outside most researchers' training (Kozak 2016). In R, these tasks are typically spread across several packages, each with its own conventions and syntax, and learning to navigate them adds effort that has little to do with the statistics. This can distract from understanding the underlying concepts and interpreting the results.

We wrote *biometryassist* in response to this, having seen through a long-running biometry training workshop series that participants consistently struggled with software mechanics as well as the statistics. The package provides a consistent interface across the key steps of an agricultural experiment, allowing users to concentrate on statistical reasoning instead of the software itself. We built it primarily for researchers, agronomists, and applied scientists who require statistically sound analyses but may not have extensive programming experience, though it applies wherever designed experiments are analysed.

## State of the field

Several R packages address individual stages of the field trial process. Experimental designs can be generated by `agricolae` (de Mendiburu 2023), `dae` (Brien 2025), and `edibble` (Tanaka

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2023); linear mixed models can be fitted with ASReml-R (The VSNi Team 2023), `lme4` (Bates et al. 2015), or `nlme` (Pinheiro et al. 2026); and post-hoc inference can be provided by `emmeans` (Lenth and Piaskowski 2026). No single tool spans the full process however, and combining several requires reconciling different syntax, conventions, and documentation standards.

We did not build *biometryassist* to replace these packages, but to wrap and extend them within one consistently documented interface. Our case for a new package rather than contributing to an existing one is straightforward: no existing package had the scope required, and the fragmentation is architectural. Improving post-hoc testing in `emmeans`, for example, would not address the design or reporting stages, nor reduce the number of packages a user must load. The documentation in `agricolae` was a particular motivation for us: its sparse and inconsistent help pages were a recurring obstacle in our workshops, and approachable documentation was one of our explicit goals.

The package also provides capabilities not available elsewhere. Buffer rows and columns can be added to generated designs for example, which is common practice in agronomic trials but unsupported by other design packages. Designs can also be exported directly to formatted Excel workbooks with a colour-coded field layout together with a spreadsheet experiment field book. The analysis functions reduce a typical analysis from around 50 lines of code across several packages to about 10, using one package, or two with ASReml-R. Fitting models remains outside the scope of *biometryassist* as the complexity of this task is better handled by the specialised packages already available. The package is fully open source and released under the MIT License.

# Software design

We organised the functions in *biometryassist* into three categories: design, for generating and randomising experimental layouts (`design()`), adding buffer plots (`add_buffers()`), and exporting them to formatted Excel workbooks (`export_design_to_excel()`); analysis, covering model checking (`resplot()`, `variogram()`), post-hoc testing and prediction (`multiple_comparisons()`, `pairwise_comparisons()`, `reference_comparisons()` and `autoplot()`); and utilities, including installation helpers (`install_asreml()`) and reporting templates (`use_template()`). This structure mirrors the stages of a typical field trial and follows the sequence of topics we cover in the workshops.

We chose function names and terminology to reflect how concepts are taught rather than how they are implemented, intending names such as `design()` and `multiple_comparisons()` to be self-explanatory to a researcher with little prior R experience.

For model fitting, *biometryassist* supports ASReml-R (The VSNi Team 2023), `lme4` (Bates et al. 2015), `nlme` (Pinheiro et al. 2026), and base `aov()`. We recommend ASReml-R for the mixed-model analyses of typical agronomic field trials, and concentrate most development effort there, but it is not a hard dependency; the package loads and functions correctly without it, and we have written it to support alternative backends where possible. This keeps *biometryassist* useful across institutional settings where a commercial ASReml-R licence may not be available.

We designed the package to extend as the R ecosystem evolves. We add support for additional model classes in response to user needs, such as the recent addition of `lme4`, `sommer` and `glmmTMB` support in `resplot()` and the `*_comparisons()` functions, and will continue to do so as new packages are developed and gain traction.

# Research impact statement

*biometryassist* began as a collection of analysis scripts we shared with workshop participants, before we formalised it as a package distributed through CRAN. Our workshop series has run for more than eight years, and the package features in each one; we encourage and support participants to apply it in their own work afterwards. Beyond the workshops, researchers have adopted it independently of our team, and it has been downloaded more than 25,000 times from CRAN as of June 2026. It has also been cited in published work spanning agricultural and other applied disciplines (Fanning et al. 2022; Reed-Métayer et al. 2025; Wenham et al. 2026; Kesser et al. 2026), demonstrating use beyond the field of agronomy for which we originally designed it.

# AI usage disclosure

We used Anthropic's Claude to suggest edits improving the clarity and flow of text we had drafted, and to assist with some recent refactoring of the package's code. In both cases we reviewed all AI-generated or AI-edited content, and confirmed the correctness of code changes with the package's existing suite of formal unit tests. We take full responsibility for the final content of both the software and this paper.

# Acknowledgements

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